

Vervus — Motion & Audio Implementation Spec

Motion and audio are specified together per game state. Web Audio API (AudioContext, decodeAudioData, createBufferSource) with HTMLAudioElement fallback.

CONSISTENCY IS CRITICAL. One motion language and one audio language across the entire platform. Motion exists to communicate state changes. Audio exists to reinforce them. Neither exists for decoration.

1. Core Rules

Motion

- One motion language across the entire platform. Lobby, game and checkout must feel identical in rhythm.
- Motion communicates state changes only. Never decorative.
- GLITCH! rounds must never be revealed through animation. SYNC and GLITCH! use identical motion.
- Silence and stillness are intentional tools. Never fill them.
- Reserve large motion for major gameplay moments. Everything else is subtle.
- Default easing: ease-out for entrances, ease-in for exits, ease-in-out for pulses.
- No animation should exceed 300ms except Kill Screen sequence and Heat Surge / LAST CHANCE overlay holds.
- Respect OS-level prefers-reduced-motion via @media (prefers-reduced-motion: reduce). When active: disable all loops, pulses and transitions. Keep instant state changes only.

Audio

- One continuous tension loop throughout the entire session. Never swap layers. Adjust parameters only.
- Multiple sounds may play simultaneously. No sound blocks another.
- All game assets — audio, images, SVGs, fonts — must be preloaded during the lobby before Start Run is possible. The Start Run button may only become active after all assets are confirmed loaded. Show a loading indicator if preloading is not yet complete.
- Sound_Click and Sound_SelectionChanged use 150ms debounce.

2. Platform

Platform sounds and motions are specified together per interaction moment.

Modal open / close

Property / Element	Duration	Values	Easing
Backdrop	200ms	opacity: 0 → 0.6	ease-out
Modal panel entrance	220ms	opacity: 0 → 1, translateY: 12px → 0	ease-out
Modal close	160ms	opacity: 1 → 0, translateY: 0 → 8px	ease-in

Bottom sheet open / close

Property / Element	Duration	Values	Easing
Sheet open	260ms	translateY: 100% → 0	ease-out (cubic-bezier 0.16, 1, 0.3, 1)

Property / Element	Duration	Values	Easing
Sheet close	200ms	translateY: 0 → 100%	ease-in
Backdrop	200ms	opacity: 0 → 0.5	ease-out

Sound

- Sound_SheetOpen plays on sheet open. Local only.

Informational overlay

Used for: reconnect screen, removed from room screen, and other informational states.

Property / Element	Duration	Values	Easing
Overlay entrance	250ms	opacity: 0 → 1	ease-out
Overlay exit	200ms	opacity: 1 → 0	ease-in

Sound

- Sound_Alert plays on overlay entrance. Local only.

FAQ accordion

Property / Element	Duration	Values	Easing
Content height	220ms	height: 0 → auto (use max-height)	ease-out
Chevron icon	220ms	rotate: 0deg → 180deg	ease-in-out

Player joined / left

Property / Element	Duration	Values	Easing
Player slot entrance	180ms	opacity: 0 → 1, scale: 0.92 → 1	ease-out
Player slot exit	150ms	opacity: 1 → 0, scale: 1 → 0.92	ease-in

Sound

- Sound_PlayerJoined broadcast to all participants on join.
- Sound_PlayerLeft broadcast to all remaining participants on leave or kick.

Primary CTA buttons — Host a Room / Join a Room / Pay

Primary action buttons on landing page and checkout. Always tappable — no conditional state.

Property / Element	Duration	Values	Easing
On tap	120ms	scale: 1 → 0.96 → 1	ease-in-out
Pay button — loading state	continuous until response	Subtle opacity pulse 1.0 → 0.7 → 1.0 while Stripe processes. Replaces tap feedback.	ease-in-out

Experience selector — landing page and lobby

Applies to the experience selector on both the landing page and the lobby. Same motion language in both contexts.

Property / Element	Duration	Values	Easing
Selected card	160ms	scale: 1 → 1.03, border emphasis as per Figma	ease-out
Deselected card	160ms	scale: 1.03 → 1, border neutral	ease-out

Sound

- Sound_SelectionChanged plays locally only for the swiping player. 150ms debounce.

Hamburger menu

Property / Element	Duration	Values	Easing
Menu open	220ms	translateY: -100% → 0, opacity: 0 → 1. Slides down from top.	ease-out
Menu close	180ms	translateY: 0 → -100%, opacity: 1 → 0	ease-in
Backdrop	200ms	opacity: 0 → 0.5	ease-out
Hamburger icon tap	120ms	scale: 1 → 0.92 → 1	ease-in-out

Lobby connection states

Each state has its own color as defined in Figma. Motion is consistent across all states — only color and text change.

Property / Element	Duration	Values	Easing
State badge entrance	180ms	opacity: 0 → 1, scale: 0.94 → 1	ease-out
State change transition	150ms	opacity: 1 → 0 then 0 → 1, color crossfade	ease-in then ease-out
Ready — on tap	140ms	scale: 1 → 1.15 → 1, color: neutral → accent	ease-in-out

States as per Figma:

- Waiting...
- Connected
- Ready
- Reconnecting...
- Host reconnecting...
- Transferring host...
- Disconnected
- Removed from room — triggers informational overlay for the removed player. See informational overlay section.

Sound

- Sound_PlayerReady broadcast to all participants on ready tap.

ROOM IS LIVE label

Always visible in the lobby while a room is active. Not a state change — a permanent ambient indicator. Gets a subtle periodic pulse to feel alive without demanding attention.

Property / Element	Duration	Values	Easing
Ambient pulse	once every 3 seconds	opacity: 1 → 0.6 → 1, scale: 1 → 1.02 → 1. Single soft pulse, not continuous breathing.	ease-in-out

Live lobby — general principle

The lobby is a live environment. Players join and leave, states change, the experience selector updates. All of these changes must feel smooth and intentional.

- No hard cuts or layout shifts in the lobby. Every change arrives with motion.
- When in doubt: fade and scale. The player should never be surprised by a sudden change.
- If multiple state changes happen simultaneously, stagger them by 60–80ms so they feel sequential rather than chaotic.
- The developer has discretion for edge cases not explicitly described here — apply the same motion language consistently.

Loading indicator — shared component

Dotted progress bar (_ _ _ _) that fills progressively and loops. Used on multiple screens:

- Guest screen while host is unlocking — 'NAME is unlocking Vervus'
- Late join screen
- Reconnecting screen
- Host reconnecting screen
- Any other waiting state

Property / Element	Duration	Values	Easing
Segment fill	sequential left to right	Each segment fills one by one. Total cycle: 1.5–2 seconds.	ease-in-out per segment
Loop reset	instant	All segments reset simultaneously at end of cycle. Immediately starts again.	—

Guest screen — host unlocking

Shown to all guests while the host completes the purchase flow.

Property / Element	Duration	Values	Easing
Screen entrance	220ms	opacity: 0 → 1, translateY: 8px → 0	ease-out
Host name	150ms, immediate	opacity: 0 → 1, scale: 0.96 → 1	ease-out
Loading indicator	starts 100ms after name	See loading indicator — shared component above.	—
Screen exit on purchase success	200ms	opacity: 1 → 0, all elements fade simultaneously	ease-in

Start Run CTA

Property / Element	Duration	Values	Easing
Button pulse	1200ms loop	scale: 1 → 1.04 → 1 — only active when session is startable (≥2 players present)	ease-in-out
On tap	120ms	scale: 1 → 0.96 → 1	ease-in-out

Start Run → Game transition

Bridge moment. All players must feel the session go live simultaneously.

Property / Element	Duration	Values	Easing
Lobby fade out	200ms	opacity: 1 → 0	ease-in
Game screen fade in	250ms	opacity: 0 → 1	ease-out
Total transition	~450ms	no gap between out and in	—

Sound

- Sound_GameTransition broadcast to all participants.
- Loop_TensionBase starts simultaneously at Standard base parameters (playbackRate 1.0, gain 0.15).

Product card selection

Property / Element	Duration	Values	Easing
Card on select	160ms	scale: 1 → 1.02, border emphasis	ease-out
CTA appearance	180ms	opacity: 0 → 1, translateY: 8px → 0	ease-out

Purchase success

Property / Element	Duration	Values	Easing
Success state	300ms	scale: 0.95 → 1, opacity: 0 → 1	ease-out
Checkmark / icon	200ms delay, 250ms	scale: 0.8 → 1	ease-out (spring-like)

Sound

- Sound_PurchaseSuccess plays. Local only — host only.

Purchase failed

Sound

- Sound_PurchaseFailed plays. Local only — host only.

Preview complete

Property / Element	Duration	Values	Easing
Overlay entrance	250ms	opacity: 0 → 1, translateY: 16px → 0	ease-out
Product card emphasis	200ms delay, 200ms	scale: 0.96 → 1	ease-out
Unlock CTA pulse	once every 4 seconds	scale: 1 → 1.03 → 1, single pulse not continuous	ease-in-out

Sound

- Sound_PreviewEnded broadcast to all participants.

3. GLiTCH! — Continuous Tension Loop

File: Loop_TensionBase.wav — seamless loop required. One loop, always on, parameters only.

State	playbackRate	gain	Notes
Standard — base	1.0	0.15	Starts with Sound_GameTransition.
Standard — combo pressure	+0.03 per 10 combo	+0.025 per 10 combo	Cap: playbackRate max 1.25, gain max 0.25.
Standard — LAST CHANCE	current + 0.15	current + 0.15	Cap: max 1.35 / 0.35. Resumes after overlay at same parameters.
Blitz — base	1.25	0.25	Starting point for Blitz mode.
Blitz — combo pressure	+0.10 per 5 combo	+0.05 per 5 combo	Cap: playbackRate max 1.75, gain max 0.60.
Heat Surge	current + 0.25	current + 0.15	Cap: max 2.00 / 0.75. Returns to pre-surge Blitz state over 400ms after overlay.

4. GLiTCH! — Game States

Round transition

Triggers every new round. Must be identical for SYNC and GLiTCH!. Must never hint at round type.

Property / Element	Duration	Values	Easing
Icon exit	80ms	opacity: 1 → 0, scale: 1 → 0.92	ease-in
Icon entrance	120ms	opacity: 0 → 1, scale: 0.92 → 1	ease-out
Total duration	~150ms	exit completes before entrance begins	—
Timer color reset	masked by icon exit	Timer resets to blue during icon exit — never visible to player. Timer ring remains in view at all times.	—

Timer

Timer ring remains visible at all times. Only the icon exits and enters each round.

Property / Element	Duration	Values	Easing
Timer ring — game start only	150ms	opacity: 0 → 1, scale: 0.96 → 1. One-time entrance at game start only.	ease-out
Timer countdown — Standard	linear, full decision window	stroke-dashoffset: full → 0. Color: blue → red, continuous linear over full window as per Figma.	linear
Timer countdown — Blitz	linear, full decision window	Identical blue → red. Glitch effects active from round start as per Figma. Visual only.	linear
Icon gradient sync — Standard / Blitz / Chaos	linear, full decision window	Icon gradient blue component transitions to red in sync with timer. White component remains — icon never fully blue/red. Always half-white as per Figma. NOTE: check asset structure with Fivecube — requires gradient layer support.	linear
Icon gradient sync — Heat Surge	during overlay only	Entire icon shifts to orange as per Figma. Transitions toward near-white overloaded tone during overlay. Resets to blue on new round after overlay exits.	linear
Timer — LAST CHANCE active	instant on trigger	Additional glitch effects on timer ring as per Figma. Visual only.	—
Timer — Heat Surge active	during heat surge only	Timer ring shifts to orange/warm as per Figma with glitch effects. Resets to blue on new round after overlay exits.	—
Timer — Chaos state 0	linear, full decision window	Identical to Standard. Clean blue → red, no corruption.	linear
Timer — Chaos states 1–9	linear, full decision window	Blue → red identical to Standard. Visual corruption	linear

Property / Element	Duration	Values	Easing
		increases per state. See Chaos section.	
Timer — Chaos state 10	linear, full decision window	Maximum corruption.	linear

Answer selection — correct

Property / Element	Duration	Values	Easing
Button feedback	100ms	scale: 1 → 0.95 → 1	ease-in-out
Correct pulse	200ms	opacity: 0 → 0.6 → 0 (green flash)	ease-out

Sound

- Sound_CorrectAnswer plays for all players at moment of result confirmation, in sync with correct visual pulse.
- Sound_CorrectAnswer plays at the same pitch and volume for every correct round including milestones. No pitch or gain escalation — tension comes from Loop_TensionBase.

Combo counter

Property / Element	Duration	Values	Easing
Every increment	120ms	scale: 1 → 1.12 → 1	ease-in-out
Milestones (10, 20, 30...)	200ms	scale: 1 → 1.25 → 1, glow intensity increases over 300ms and settles	ease-in-out

Heat Surge

Full-screen state overlay. Covers the entire game screen. Must feel like maximum overdrive — harder and more violent than LAST CHANCE.

Property / Element	Duration	Values	Easing
Screen impact on entry	instant	Single hard flash — scale: 1.08 → 1 with opacity spike. Violent and sudden.	ease-out
Full-screen overlay entrance	100ms	opacity: 0 → 1, scale: 1.06 → 1	ease-out
Overlay visible	~500ms	Static. Player sees full Heat Surge state screen.	—
Overlay exit	120ms	opacity: 1 → 0, scale: 1 → 0.97	ease-in
Return to game	instant after exit	New round begins. Timer resets to blue masked by overlay exit.	—

Property / Element	Duration	Values	Easing
Audio layer return	400ms	playbackRate and gain ease back to pre-surge Blitz state.	ease-in-out
Background intensification	300ms	Color/brightness shift to Heat Surge state during overlay.	ease-out
Active icon intensification	300ms	Glow increase, scale 1 → 1.04. Applies to active icon regardless of type.	ease-out

Sound

1. All active audio nodes stop instantly when overlay enters.
2. Sound_HeatSurgeWarning plays. All other sounds stop — Sound_HeatSurgeWarning continues playing through overlay and into new round regardless of overlay duration.
3. When overlay exits: Loop_TensionBase resumes at Heat Surge parameters (current Blitz + 0.25 / + 0.15, capped at 2.00 / 0.75).
4. New round begins. After 400ms ease-in-out, loop returns to pre-surge Blitz parameters for current combo.

LAST CHANCE

Full-screen state overlay. Freeze is absolute — everything stops instantly before the overlay appears.

Property / Element	Duration	Values	Easing
Freeze — all animation	instant (0ms)	All running animations stop immediately.	—
Freeze hold	500ms	Complete stillness. Nothing moves.	—
Screen impact on entry	instant	Single hard flash — scale: 1.04 → 1 with opacity spike. Less violent than Heat Surge.	ease-out
SAVE IT overlay entrance	120ms	opacity: 0 → 1, scale: 0.96 → 1	ease-out
Overlay visible	~500ms	Static. Player sees full LAST CHANCE state screen.	—
Overlay exit	100ms	opacity: 1 → 0	ease-in
Return to game	instant after exit	Same round begins from start. Timer resets to blue masked by overlay exit. Combo intact.	—

Sound

5. All active audio nodes stop instantly at freeze moment — simultaneous with animation freeze.
6. Sound_LastChanceFreeze plays.
7. When overlay exits: Loop_TensionBase resumes immediately at same parameters as before freeze.

8. Same round replays from start. Loop continues uninterrupted.

Fail state

Abrupt. Final. The stillness is intentional — do not fill it.

Property / Element	Duration	Values	Easing
All animation — stop	<code>instant (0ms)</code>	Everything cuts immediately.	—
Screen impact flash	<code>80ms</code>	opacity: 0 → 0.7 → 0 (white flash)	ease-out
Stillness	<code>300ms</code>	Nothing moves — intentional gap before Kill Screen.	—

Sound — Fail Sequence

9. Stop all active audio nodes immediately.
10. Play Sound_FailImpact.
11. 300ms intentional silence — do not fill.
12. Play Sound_KillScreenSting and reveal Kill Screen simultaneously.

Kill Screen

Enters immediately after the 300ms stillness. Sharp, not soft. Elements reveal in sequence — never all at once.

Property / Element	Duration	Values	Easing
Kill screen reveal	<code>200ms</code>	opacity: 0 → 1	ease-out
VERVUS + GLITCH! logos	<code>150ms, immediate</code>	opacity: 0 → 1	ease-out
Combo count-up	<code>starts 100ms after logos</code>	0 to final combo. Target total: 1.0–1.5 sec regardless of combo size. Developer calculates ms per digit dynamically.	linear
WHO BROKE THE RUN container	<code>180ms, after count-up completes</code>	opacity: 0 → 1, translateY: 6px → 0	ease-out
Guilty player name — emphasis	<code>200ms delay after container, then 300ms</code>	scale: 1 → 1.08 → 1, red glow pulse. Single pulse — not continuous. Draws the eye after the container has settled.	ease-in-out
REALITY CHECK container	<code>180ms, 150ms after WHO BROKE THE RUN</code>	opacity: 0 → 1, translateY: 6px → 0	ease-out
EXPERIENCE container	<code>150ms, 100ms after Reality Check</code>	opacity: 0 → 1	ease-out
PLAYERS container	<code>150ms, 100ms</code>	opacity: 0 → 1	ease-out

Property / Element	Duration	Values	Easing
	after Experience		
Another Run CTA	150ms, 200ms after Players	opacity: 0 → 1, scale: 0.96 → 1	ease-out
Share card	150ms, 100ms after CTA	opacity: 0 → 1, translateY: 6px → 0	ease-out

Sound

- Sound_ResultsTick plays once per digit during combo count-up. Constant pitch and volume throughout.

Chaos — corruption states

Rule: corrupt the presentation, never the gameplay. The player must never lose because the timer visually lied about remaining time.

Audio distortion increases in sync with visual corruption per combo band. Code-driven.

State	Corruption	Visual + Audio description
0	None	Identical to Standard. Clean visuals and audio. No corruption.
1–2	Minimal	Occasional single-frame flicker on timer. Subtle segment opacity variation. Audio: near-zero distortion.
3–4	Light	Brief flickering on 1–2 timer segments. Small opacity drops. Audio: faint distortion layer begins.
5–6	Moderate	Temporary segment loss on 1–3 timer segments. Brief visual desync. Cracks appear on icon. Audio: distortion audible but not dominant.
7–8	Heavy	Ring fragmentation visible. Multiple simultaneous corruptions. Aggressive icon cracking. Audio: distortion prominent.
9	Severe	Near-collapse. Ring appears broken. Aggressive fragmentation and distortion. Audio: heavy distortion.
10	Maximum	Timer visually falling apart. Segments scatter. Icon may partially fragment. Must remain readable enough to feel dangerous. Audio: maximum distortion within fairness limits.

5. Admin Panel — Audio Settings

Expose per sound at minimum:

- Volume
- Playback Rate
- Detune
- Fade In
- Fade Out
- Chaos distortion per combo band.

6. Global Rules

Motion

- All entrance animations use ease-out.
- All exit animations use ease-in.

- All pulses and loops use ease-in-out.
- SYNC and GLITCH! rounds are visually identical until result is revealed.
- The 300ms stillness after Fail State and the 500ms freeze during LAST CHANCE are never shortened or filled.
- Animations that repeat must never become visually noisy. If in doubt, reduce intensity.

Audio

- Heat Surge and LAST CHANCE both cut all audio instantly on entry. Loop resumes after overlay exits.
- Fail State audio cut is permanent. Loop does not resume. Kill Screen has its own sounds only.
- All sounds that continue playing after an overlay (e.g. Sound_HeatSurgeWarning) are never stopped by the overlay — they play through regardless of overlay duration.
- Loop_TensionBase must support seamless looping via source.loopStart and source.loopEnd.
- Sound_Click uses 150ms debounce — must not trigger during rapid swipe interactions.
- Sound_SelectionChanged uses 150ms debounce.
- iOS Safari requires AudioContext to be resumed on explicit user interaction. AudioContext is activated on the Start Run tap. This is sufficient — no re-activation needed after audio cuts (Heat Surge, LAST CHANCE, Fail State) as these stop nodes but not the AudioContext itself.
- If the user leaves the tab and returns, iOS Safari may suspend the AudioContext. Listen for the visibilitychange event and call AudioContext.resume() when the document becomes visible again. No new user interaction required as the context was already activated earlier in the session.